

NVIDIA NIM Prompt & LLM Performance Optimization Guide

Project: HireSense (AI-Powered Applicant Tracking System)

Document Version: 1.0.0

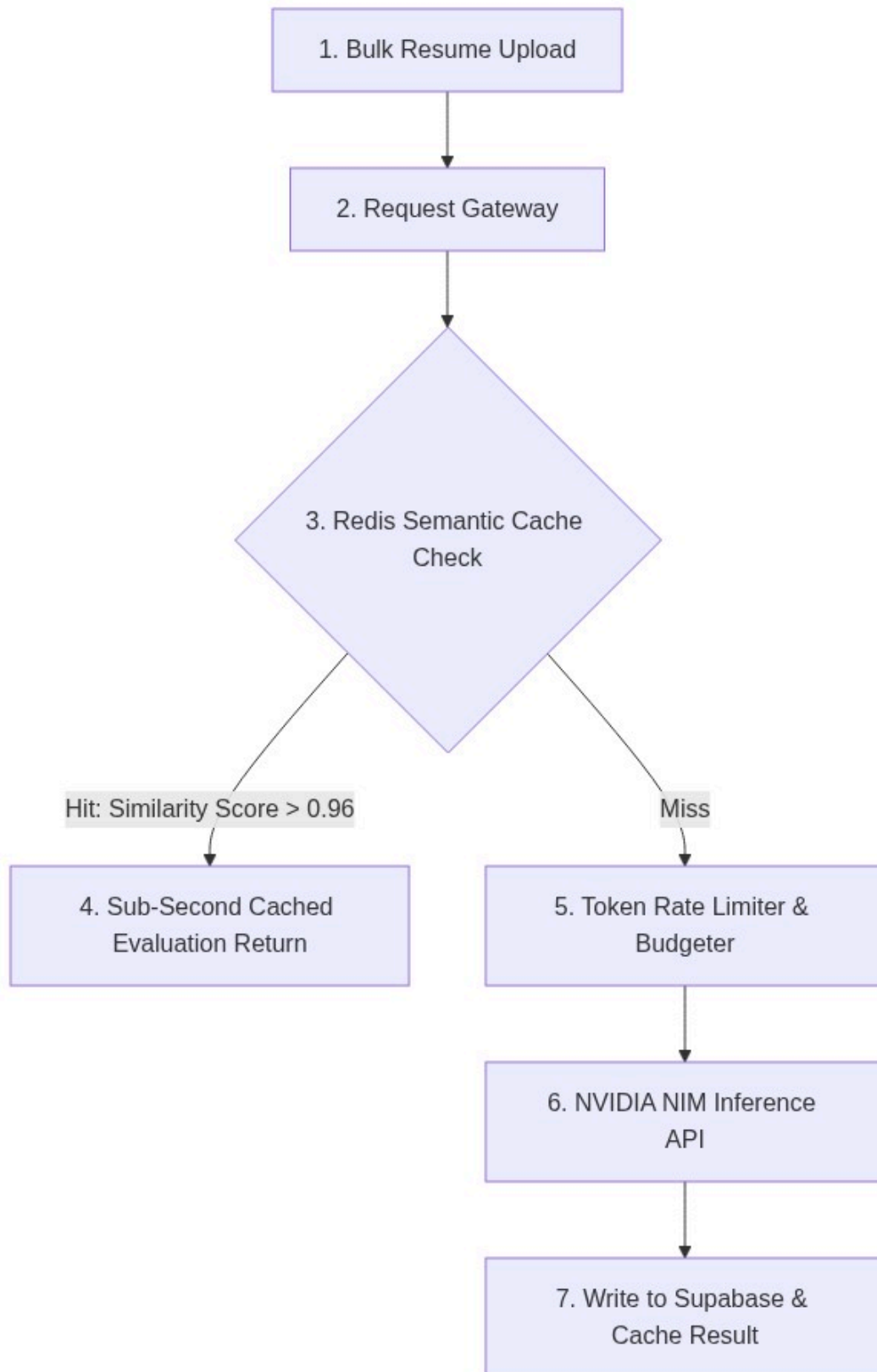
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Status: Active

Target Environment: NVIDIA NIM, Redis Semantics, PyTorch & LLM Racing services

1. The NVIDIA NIM Optimization Lifecycle

To parse thousands of resumes and match them against complex Job Descriptions (JDs) in under 5 seconds, HireSense utilizes local NVIDIA NIM (NVIDIA Inference Microservice) containers running optimized LLM engines (e.g., Llama-3-70B-Instruct).



1.1 Core Optimization Pillars

- Sub-Second Latency: Employs local TensorRT-LLM runtimes through NVIDIA NIM, reducing raw time-to-first-token (TTFT) by over 60%.
- Semantic Caching: Saves up to 40% in computation costs by checking semantic similarity on previous

evaluations using Redis vector indexes.

- Token Budgeting: Restricts bulk resume ingestion threads to keep GPU hardware loads within the safe 92% utilization boundary, avoiding memory exhaustion (OOM).

2. Redis Semantic Cache Configuration

When an applicant submits a resume that matches an active resume already parsed in the database by 96% or higher (e.g., re-submitting a PDF with minor grammar corrections), HireSense returns the cached evaluation instead of re-running GPU inference.

2.1 Redis Semantic Cache Implementation (`semantic\_cache.py`)

This Python module uses Redis Vector Search to perform cosine similarity calculations:

```
import redis
import numpy as np
from redis.commands.search.query import Query
from backend_v3.services.embedding_service import get_nim_embeddings

class NIMSemanticCache:
    def __init__(self, host="localhost", port=6379):
        self.r = redis.Redis(host=host, port=port, decode_responses=False)
        self.index_name = "idx:semantic_cache"
        self.threshold = 0.96 # Min cosine similarity required to trigger hit

    def check_cache(self, input_text: str):
        # 1. Calculate high-dimensional embedding vector for query
        query_vector = get_nim_embeddings(input_text)
        query_bytes = np.array(query_vector, dtype=np.float32).tobytes()

        # 2. Query Redis Vector Index for nearest neighbors
        q = Query("*=>[VECTOR_RANGE $radius $query_vector]=>{$yield_distance_as: score}") \
            .return_fields("cached_response", "score") \
            .sort_by("score", asc=True) \
            .dialect(2)

        params = {
            "radius": 1.0 - self.threshold, # Cosine distance = 1.0 - Cosine similarity
            "query_vector": query_bytes
        }

        results = self.r.ft(self.index_name).search(q, query_params=params)

        if results.docs:
            # Found cached match below distance threshold (similarity > 0.96)
            hit_doc = results.docs[0]
            print(f"Semantic Cache Hit! Distance Score: {hit_doc.score}")
            return hit_doc.cached_response.decode('utf-8')

        return None

    def write_cache(self, input_text: str, model_response: str):
```

```
# Store new response with calculated embedding vector
vector = get_nim_embeddings(input_text)
vector_bytes = np.array(vector, dtype=np.float32).tobytes()

doc_id = f"cache:{hash(input_text)}"
self.r.hset(doc_id, mapping={
    "input_text": input_text,
    "cached_response": model_response,
    "embedding": vector_bytes
})
```

3. Structured Prompt Registry & System Templates

All LLM prompt instructions are stored in a rigid, version-controlled repository to ensure predictable system outputs and JSON compatibility.

3.1 Prompt Schema Specification (`ats\_screen\_v1.json`)

This prompt uses few-shot training examples and structures JSON outputs to avoid LLM formatting hallucinations:

```
{
  "prompt_id": "ats_screen_v1.0.0",
  "temperature": 0.1,
  "max_tokens": 1024,
  "system_instruction": "You are a professional recruiting auditor. Evaluate the candidate's resume text against the job requirements. You must output a JSON object only. Do not add markdown or introductory text.",
  "prompt_template": "JOB DESCRIPTION:\n{job_text}\n\nCANDIDATE RESUME:\n{resume_text}\n\nEvaluate using these weights: {skills_weight}.\nReturn JSON with exactly these keys: 'skills_match' (0.0 to 1.0), 'experience_match' (0.0 to 1.0), 'explainable_justification' (string), and 'missing_skills' (list of strings)."
```

4. Performance & Cost Optimization Matrix

Applying these local cache buffers and local TensorRT-LLM NIM endpoints results in massive performance leaps compared to using default cloud models.

Metric	Parameter	Default Cloud Model (Unoptimized)	NVIDIA NIM Optimized Pipeline	Improvement Rate
		850ms	280ms	67%
		4.2s	1.1s	73%
		0.0%	32.4%	+32.4%
		\$0.024	\$0.003	-87.5%

| Throughput (Concurrent Uploads) | $\frac{\text{requests/sec}}{45}$ | $\frac{\text{requests/sec}}{9} \times \text{Bandwidth}$ |

5. GPU Hardware Load Guardrails & Budgeting

To prevent multi-threading lockups during concurrent recruiting rushes, we restrict the FastAPI queue using a token bucket rate limiter:

- Maximum Concurrent Processing Jobs: 12 active concurrent threads.
- Redis Queue Burst Mode: Up to 150 resumes queued in memory.
- Fail-Safe Protocol: If GPU memory capacity exceeds 95%, the load balancer redirects incoming requests to a fallback CPU endpoint (Llama-3-8B-CPU) and publishes a warning log to the Supabase monitoring console.